

SEUNGHEON DOH

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<https://seungheondoh.github.io/>

RESEARCH INTEREST

Music Information Retrieval, Language Models, Multimodal Representation Learning

My research focuses on developing music intelligence for understanding, retrieval, generation, and post-production tasks. This work is structured around three interconnected research directions. First, I investigate **multi-modal representation learning** methods that establish semantic correspondences between music and complementary modalities including natural language, visual content, and other media formats. These representations serve as foundational components for transfer learning in music-related applications. Second, I explore **multi-modal large language models** (MLLMs) for music applications, with a focus on three key subtopics: reasoning, chain-of-thought processes, and tool calling, leveraging the intermediate generation capabilities of language models. Third, I design and evaluate **conversational interfaces** through real-world applications, focusing on user experience, usability, and satisfaction in diverse music-related scenarios, ensuring that the developed technologies meet practical needs and deliver measurable value to end users.

TUTORIAL

Connecting Music Audio and Natural Language

Seungheon Doh, Ilaria Manco, Zachary Novack, Jong Wook Kim and Ke Chen
The 25nd International Society for Music Information Retrieval Conference (ISMIR), 2024

SELECTED PUBLICATIONS

TalkPlay-Tools: Conversational Music Recommendation with LLM Tool Calling

Seungheon Doh, Keunwoo Choi, and Juhan Nam
The Workshop on AI for Music, Neural Information Processing Systems, (NeurIPS-AI4Music), 2025 RecSys

TalkPlay: Multimodal Music Recommendation with Large Language Models

Seungheon Doh, Keunwoo Choi, and Juhan Nam
IEEE Transactions on Audio, Speech and Language Processing (IEEE-TASLP - Under Review), 2025 RecSys

TalkPlayData-2: An Agentic Synthetic Data Pipeline for Multimodal Conversational Music Recommendation

Keunwoo Choi, Seungheon Doh, and Juhan Nam
The International Conference on Learning Representations (ICLR-Under Review), 2025 RecSys

LLM2Fx-Tools: Tool Calling for Music Post-Production

Seungheon Doh, Junghyun Koo, Marco A. Martínez-Ramírez, Woosung Choi, Wei-Hsiang Liao, Qiyu Wu, Juhan Nam, and Yuki Mitsufuji
The International Conference on Learning Representations (ICLR-Under Review), 2025 Production

LP-MusicCaps: LLM-based Pseudo Music Captioning

Seungheon Doh, Keunwoo Choi, Jongpil Lee, Juhan Nam
The International Society for Music Information Retrieval Conference (ISMIR), 2023
(Best Paper Award Nomination - Long Talk) Understanding

More Publications @ <https://scholar.google.com/citations?user=MCKggcgAAAAJ> (Total Citation: 573)

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) Ph.D in Graduate School of Culture Technology Music and Audio Computing Lab (Advisor: Juhan Nam)	2021 - 2025
Korea Advanced Institute of Science and Technology (KAIST) MSc. in Graduate School of Culture Technology Music and Audio Computing Lab (Advisor: Juhan Nam)	2019 - 2021
Ulsan National Institute of Science and Technology (UNIST) B.S. in School of Business administration & Industrial Design	2014 - 2019

EXPERIENCE

SonyAI Tokyo, Japan Research Intern (Advisor: Junghyun Koo, Marco A. Martínez-Ramírez) Topic: Large Language Model for Music Production (Multimodal LLM)	Apr 2025 - Nov 2025
Adobe Research San Francisco, United States Research Intern (Advisor: Nick Bryan, Ge Zhu, Zhepei Wang) Topic: Large Language Model for Music Generation (Multimodal LLM)	Jun 2024 - Sep 2024
Chart Metric Remote Research Intern (Advisor: Keunwoo Choi) Topic: Artist Similarity Retrieval System (Artist-to-Artist)	Dec 2023 - Mar 2024
Naver Corp. Seongnam, South Korea Research Intern in Now AI Team (Advisor: Jeong Choi) Topic: Multimodal Video Recommendation System (User-to-Item)	Dec 2022 - Mar 2023
NYU, Computational Intelligence, Learning, Vision, and Robotics (CILVR) New York, United States Visiting Student (Advisor: Kyunghyun Cho) Topic: Self-supervised Learning Model for Music	Jun 2022 - Aug 2022
ByteDance/Tiktok Remote Research Intern in Speech, Audio & Music Intelligence Team (Advisor: Keunwoo Choi, Minz Won) Topic: Multimodal Music Retrieval System (Text-to-Music, Speech-to-Music)	Jul 2021 – Jan 2022
NYU, Music and Audio Research Laboratory (MARL) New York, United States Visiting Student (Advisor: TaeHong Park) Topic: Word Embedding for Music Annotation and Retrieval	Dec 2019 - Feb 2020

OPEN-SOURCE SOFTWARE

- **LP-MusicCaps:** Music Captioning System (Released 2023, 300+ stars):
- **TTMR++:** Text-to-Music Retrieval System (Released 2025, 40+ stars):
- **TTMR:** Text-to-Music Retrieval System (Released 2022, 110+ stars):

ADVISING

Have advised research internship students. Academic papers have been presented in top-tier conferences and workshops (ISMIR, HCMIR, NLP4MusA).

- Daeyong Kwon, Retrieval Augmented Generation for Music Question Answering (2025)
- Hayeon Bang, Audio-Text-Midi Joint Embedding Space (2024)
- Minhee Lee, Audio-Text Joint Embedding Space (2023)
- Yoonjin Chung, Music-to-Image Generation (2022)
- Youngjune Choi, Hierarchical Audio Few-shot Learning (2022)
- Haven Kim, Playlist Title Generation (2022)
- Junwon Lee, Playlist Title Generation (2021)
- Hounsung Kim, Playlist Continuation (2020)
- Nabin Kim, Music Emotion Recognition (2020)

TEACHING ACTIVITY

- TA, GCT731 Topics in Music Technology: Generative AI for Music (Mar 2023)
- TA, GCT634 Musical Applications of Machine Learning (Sep 2022)

- TA, GCT634 Musical Applications of Machine Learning (Sep 2021)
- TA, GCT731 Topics in Music Technology: Cognitive Science of Music (Sep-2020)
- TA, GCT576 Social Computing (Sep 2019)

TALK

- Invited Talk, Connecting Music and Large Language Model, Mix.Audio
Feb 2025, Host: Jongpil Lee
- Invited Talk, Connecting Music and Large Language Model, TikTok/ByteDance
Aug 2024, Host: Ju-Chiang Wang
- Invited Talk, LLM-Powered Music Annotation and Retrieval, KAIST
Jun 2024, Host: Juhan Nam
- Invited Talk, Text-based Music Annotation and Retrieval, UC San-Diego
Mar 2024, Host: Hao-Wen Dong
- Invited Talk, Multimodal Music Retrieval for Listener and Contents Creator, Seoul National Univ.
Feb 2023, Host: Junghyun Koo
- Invited Talk, Music Informational Retrieval with Natural Language Processing, Yonsei Univ.
Dec 2022, Host: Saebyul Park

SERVICE

- Organizing Committee, 4th Workshop on Natural Language Processing for Music Audio, ECAL 2026
- Organizing Committee, Newcomer Initiative Chair, ISMIR 2025
- Organizing Committee, 1st Workshop on Large Language Models for Music & Audio, ISMIR 2025
- Organizing Committee, 3rd Workshop on Natural Language Processing for Music Audio, ISMIR 2024
- Reviewer, ISMIR, ICASSP, ICLR, ICML, NeurIPS, IEEE TASLP

LANGUAGES & SKILL & INTERESTS

- English(fluent), Korean(native)
- Python (Pytorch, Tensorflow), HTML/CSS, React.js, C++
- Media Art, Graphic Design, Prototyping & Making, Piano, Movie, Crossfit, Lifting

REFERENCES

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| • Juhan Nam, Professor, KAIST | juhan.nam@gmail.com |
| • Keunwoo Choi, Senior Research Director, Gaudio Lab | gnuchoi@gmail.com |
| • Minz Won, Research Scientist, Suno | minz.s.won@gmail.com |
| • Jongpil Lee, CEO, Mix.Audio | jongpillee@neutune.com |